NOTRE DAME UNIVERSITY BANGLADESH

CSE LAB Report-01

Submitted by:

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Batch: CSE-20

Date: 15-07-23

Course Title: Data Structure Lab

Course Code: CSE-2104

Lab Task Topic: Array print, Index, Sum, Average, Intersection.

Submitted To:

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Problem No. 01

Name of the problem: Write a program in C to store N numbers of elements in an array and print it.

Solution:

```
//Lab Task 1 Question no. 1
//print array numbers
#include<stdio.h>
int main()
{
    printf("Enter no. of elements: ");
    int n,i;
    scanf("%d",&n);
    int a[n];
    for(i=0;i<n;i++)
                scanf("%d",&a[i]);
        printf("\nArray is:\n");
    for(i=0;i<n;i++)
    printf("\ta[%d]=%d\n",i,a[i]);
    return 0;
}
```

Output:

```
C:\Users\istia\Downloads\20_05_q1.exe
Enter no. of elements: 5
12 23 34 45 54

Array is:
        a[0]=12
        a[1]=23
        a[2]=34
        a[3]=45
        a[4]=54

Process returned 0 (0x0)   execution time : 20.401 s
Press any key to continue.
```

Problem No. 02

Name of the problem: Write a program in C to find out the even values of an array.

Solution:

```
//Lab Task 1 Question no. 2
//Find out the even values of an array
#include<stdio.h>
int main()
{
    printf("Enter no. of elements: ");
    int n,i;
    scanf("%d",&n);
    int a[n];
    for(i=0; i<n; i++)
        scanf("%d",&a[i]);
    printf("\nArray is: \n");
    for(i=0;i<n;i++){
    if(a[i]%2==0)
    printf("\tA[%d]=%d\n",i,a[i]);
    }
    return 0;
}
```

Output:

```
C:\Users\istia\Downloads\20_05_q2.exe
Enter no. of elements: 6
12 11 34 55 68 90

Array is:
        A[0]=12
        A[2]=34
        A[4]=68
        A[5]=90

Process returned 0 (0x0)   execution time : 22.215 s
Press any key to continue.
```

Problem No. 03

Name of the problem: Write a program in C to find out the values of the odd indexes of N numbers of elements of an array.

Solution:

```
//Lab Task 1 Question no. 3
//find out the values of the odd indexes
#include<stdio.h>
int main()
{
    printf("Enter no. of elements: ");
    int n,i;
    scanf("%d",&n);
    int a[n];
    for(i=0;i<n;i++)
                scanf("%d",&a[i]);
    printf("\nArray is: \n");
    for(i=0;i<n;i++){
    if(i%2!=0)
    printf("\tA[%d]=%d\n\n",i,a[i]);
    }
    return 0;
}
```

Output:

```
C:\Users\istia\Downloads\20_05_q3.exe
Enter no. of elements: 7
12 23 34 45 56 67 69

Array is:
        A[1]=23

        A[3]=45

        A[5]=67


Process returned 0 (0x0)   execution time : 20.030 s
Press any key to continue.
```

Problem No. 04

Name of the problem: **Write a program in C to find the sum of all elements of the array of N elements.**

Solution:

```
//Lab Task 1 Question no. 4
//find the sum of all elements
#include<stdio.h>
int main()
{
    printf("Enter no. of elements: ");
    int n,i;
    scanf("%d",&n);
    float a[n],sum=0.0;
    for(i=0; i<n; i++)
        scanf("%f",&a[i]);
    for(i=0; i<n; i++)
    {
        sum=a[i]+sum;
    }
    printf("\nSum of %d elements is %f",n,sum);
    return 0;
}
```

Output:

```
C:\Users\istia\Downloads\20_05_q4.exe
Enter no. of elements: 4
12 23 34 45

Sum of 4 elements is 114.000000
Process returned 0 (0x0)   execution time : 16.592 s
Press any key to continue.
```

Problem No. 05

Name of the problem: Write a program in C to find the average of all elements of the array.

Solution:

```
//Lab Task 1 Question no. 5
//find the average of all elements
#include<stdio.h>
int main()
{
    printf("Enter no. of elements: ");
    int n,i;
    scanf("%d",&n);
    float a[n],sum=0.0,avg;
    for(i=0; i<n; i++)
        scanf("%f",&a[i]);
    for(i=0; i<n; i++)
    {
        sum=a[i]+sum;
    }
    avg=sum/n;
    printf("\nAverage of %d elements is %f",n,avg);
    return 0;}
```

Output:

```
C:\Users\istia\Downloads\20_05_q5.exe
Enter no. of elements: 4
12 23 34 45

Average of 4 elements is 28.500000
Process returned 0 (0x0)   execution time : 24.391 s
Press any key to continue.
```

Problem No. 06

Name of the problem: Find out the intersection values of 2 arrays.

Solution:

```
//Lab Task 1 Question no. 6
//intersection values of 2 arrays.
#include<stdio.h>
int main()
{
    printf("Enter no. of elements of A: ");
    int n,i,m,j;
    scanf("%d",&n);
    int A[n];
    for(i=0; i<n; i++)
        scanf("%d",&A[i]);
    printf("Enter no. of elements of B: ");
    scanf("%d",&m);
    int B[m];
    for(i=0; i<m; i++)
        scanf("%d",&B[i]);
    printf("Intersection : ");
    for(i=0; i<n; i++)
    {
```

```
        for(j=0; j<m; j++)
        {
            if(A[i]==B[j])
            printf("%d ",A[i]);
            break;
        }
    }
    return 0;
}
```

Output:

```
C:\Users\istia\Downloads\20_05_q6.exe
Enter no. of elements of A: 3
12 23 34
Enter no. of elements of B: 4
23 35 34 46
Intersection : 23 34
Process returned 0 (0x0)   execution time : 25.896 s
Press any key to continue.
```